

# January 2026 CSE 406: Cyber Security Sessional

## Online 1: Buffer Overflow

Section: A1

Time: 25 minutes

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### Objective

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You are given a vulnerable C program (`target.c`) that uses global buffers for authentication and to detect memory corruption. Your task is to exploit a **two-stage buffer overflow** in the BSS segment to bypass both the canary and the authentication check.

### Environment Setup

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```
seed@ubuntu:~$  
sudo ln -sf /bin/zsh /bin/sh
```

**IMPORTANT:** Note that we are **NOT** turning ASLR off.

### Student ID Configuration

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Open `target.c` and replace `<LAST_3_DIGITS_OF_YOUR_ID>` in the `#define STUDENT_ID` line with the **last 3 digits** of your Student ID. Do not keep leading zeroes.

For example, if your Student ID is 2105085, set:

```
#define STUDENT_ID 85
```

### Compilation

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```
seed@ubuntu:~$  
gcc -m32 -o target -fno-stack-protector target.c  
sudo chown root target  
sudo chmod 4755 target  
touch badfile  
gcc -m32 -o target_dbg -g -fno-stack-protector target.c
```

### Task

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Write a Python exploit script (`exploit.py`) that generates a `badfile` such that when the program is run, it prints:

```
seed@ubuntu:~$
```

```
Access Granted - Welcome, Administrator!
```

And spawns a root shell.

## Restrictions

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1. **DO NOT** turn ASLR off
2. **Do NOT modify** `target.c` (except for setting your `STUDENT_ID`)
3. Your exploit must work with the specific buffer sizes and canary value determined by **your** Student ID

## Submission

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1. Create a folder with your student ID as name (i.e., 2105185)
2. Put `target.c` and `exploit.py` inside it.
3. Zip the folder and upload it.